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Towards Predicting Attention And Workload During Math Problem Solving

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Blind

Abstract. Gifted students are characterized by a low level of attention and workload. Thus, it is very important to detect the variation of these values in real time when children are solving problems. A low value of workload or attention could be an indicator that the child fails the task and we should in this case intervene to avoid this problem. In this paper, we conducted a preliminary study in order to detect when children have a low values of attention or workload. A sample of 17 pupils participated in this study by solving math problems in an environment called Netmath. The EEG signal data collected from the experiment was used to train an LSTM to predict two mental states (attention and workload) in real time, when solving math problem. First results show that it is possible to predict these values in real time and the accuracy of the prediction is slightly above the random model. This pilot research provide some insight to the hypothesis that we can predict those variables in real time, which might be useful to intelligent tutor and to detect gifted children.

Keywords: Electroencephalography (EEG) prediction · LSTM · Attention · Workload.

1 Introduction

It has been shown that gifted students have low values of attention and cognitive workload when solving problems [7]. Hence, if we are able to predict the attention and the workload of a student in a real time setting during problem solving, we will be able to predict when he/she will have great chance to succeed, and also be able to predict if he/she is gifted. While existing studies aim at classifying mental states in real time, predicting them have not been explore yet. Therefore we conducted a pilot study where we have developed a LSTM [8] model for predicting student's attention and cognitive workload when solving problems. The goal is to reach a model that would be able to accurately predict what the mental sate of the learner would be at time $t+1$ knowing what it was at time $t, t-1, ..., t-T$ where T represents the timestep of the model. Such a model can be useful in an intelligent tutor for helping infer student's mental states and decide accordingly what and how to tutor at each moment.

The paper is organized as follows. Section 2 presents related work. Section 3 describes the model, the experiment settings following by the results and discussions. Finally, Section 4 presents our conclusions.

2 Related Work

2.1 Gifted Students

Historically, there are many definitions and conceptualizations of gifted and talented students [6,10,14]. Some authors allege that the high intellectual potential is innate (genetically present) and others that it represents the result of training or development of abilities or capacities of the child. Intellectual assessment or intelligence quotient remains an important indicator of giftedness. Two essential models are used to define giftedness [12], the one of Renzulli [10] and the other of Sternberg [11]. According to Renzulli [10], there are two types of gifted students: the first type corresponds to those with high academic potential. The second type corresponds to those with high creative potential. It proposes three components of skills to characterize the behavior of gifted children (intelligence, creativity and implication). These components interact. Sternberg [12] described a model of five criteria (excellence in one area relative to other people, scarcity of the level reached against peers, potential to produce something, ability to demonstrate skills with a valid assessment, and relative value of the skill for society).

2.2 Capturing Mental states

The EEG signal is a voltage signal that can be measured on the surface of the scalp, arising from large areas of coordinated neural activity manifested as synchronization (groups of neurons firing at the same rate) [4]. EEG signal can measurably detect variation in the neural activity. Attention and cognitive workload are mental states that can be tracked by a EEG headset. Many studies have used those mental states to predict reading behavior in a tutor [4], to predict performance level during cognitive task [3], to predict math problem solving [1] or to monitor indexes of alertness, cognition, and memory [2]. Berka and her team [2] used indexes extracted from EEG in order to study the engagement and workload mental states , Chaouachi et al. [5] integrated these two mental states in their system, Mentor. This system used some rules in order to maintain students in a positive mental state while learning, and reacted each time on selecting the appropriate next activity to present to the learner. This initial results appear to suggest that EEG might be a valuable technology for directly assessing a students level of cognitive effort. However, none of the previous works led to a model for predicting the next mental state of the learner. Yet such information could be of interest if we want to prevent undesirable mental states that might affect the learning process. The ideal predictive model should preserve the temporal and sequential nature data making the LSTM the best potential candidate.

2.3 Using Long Short Term Memory (LSTM) for predicting Attention and the Workload

LSTM has became a core component of neural Natural Language Processing (NLP) [13]. The LSTM architecture [8] can learn long-term dependencies using

a memory cell that is able to preserve states over long periods of time. While numerous LSTM variants have been proposed, here we used the version proposed by Zaremba [16]. It is a model able to extract sequence information from data. It is suitable for problems where the temporal prediction is important as predicting attention in real time based on past values. Given a sequence $x_1, ..., x_T$ where x_i represents the concatenation of data depicting the mental states at time $t = i$s and $T = 20s$ (time-step of our model), the LSTM processes each x at a time, keeping track of cell and state vectors (c_t, h_t) which are computed as a function of x_t, c_{t-1} [9]. In the next section, we describe how the model has been implemented in this study.

3 Experiments

We conducted an experiment where we asked elementary school students (4th and 5th grades) to solve the selected tasks from NetMath [1] environment which is a web application to support learning mathematics for primary and secondary students (from 3rd primary grade to 4th secondary grade). The proposed tasks were designed for higher-level students (6th grade). 17 students (10 F, 7 M) voluntarily participated in this study. Students are aged between 9 and 11 years (M = 10.05; SD = 0.42). We choose a total of 10 tasks from the platform. These tasks are divided into three levels of difficulty: easy, medium and hard. During the fulfillment of the tasks, we collected real time data from Neeuro Senzeband non-invasive EEG headset. This headset allows us to obtain EEG raw data from 4 channels and three mental states measures (Attention, Workload and Relaxation).

3.1 Dataset

Each of the 17 participants spent around 30 minutes to solve all the problems. For each of these participants we have gathered their attention, cognitive workload, relaxation, both hemispheres of the brain left/right, center-left and center-right at each second. For the prediction of attention and workload, we have defined a step of 20 seconds which defined the timestep for the LSTM model. The dataset was thus divided into input data X and output data Y where each line of X represents events that happened 20 seconds before and each line of Y is a label which is the event that happened at the 21st second. This configuration gave us around 20000 data where 75% where used for training and 25% for test. Fig. 1 shows an excerpt of the dataset.

3.2 The Model

Initially we trained a regression model (LSTM with mean squared error) whose objective was to predict the approximate value of attention/workload but this

[1] https://www.netmath.ca/fr-qc/

Time	Attention	Workload	Relaxation	Left	CenterLeft	CenterRight	Right
16:13:29 PM	0,998691	0,3732375	0,1199541	-6	-13	-8	-2
16:13:30 PM	0,01	0,01	0,07488824	8	10	68	15
16:13:32 PM	0,89289	0,01	0,01517821	-14	11	113	29
16:13:32 PM	0,89692	0,01	0,05044472	-6	-7	6	10
16:13:33 PM	0,904556	-0,07081398	0,07155754	15	-6	-46	-5
16:13:34 PM	1	0,2673379	0,06456525	13	11	-9	9
16:13:35 PM	0,968494	0,01	0,0776386	8	4	-2	6
16:13:36 PM	0,969414	0,5121103	0,09333985	50	9	9	16
16:13:37 PM	0,359994	0,01	0,0469471	47	-2	-21	9
16:13:38 PM	0,724037	0,276055	0,09904281	39	4	3	13
16:13:39 PM	0,970849	0,333482	0,1598548	19	3	6	9

Fig. 1. Excerpt of the raw EEG data.

first model gave us poor results. We have therefore turned the problem into a classification problem where we predict whether the value of the attention is greater than a threshold value or not. The best threshold value found is 0.6 (0.02 for the cognitive workload). So the designed model is able to predict whether attention will be high (> 0.6) or low ($<=0.6$). For example, if the value to predict is less than 0.6, then the label is encoded as [1,0] else the label is encoded as [0,1]. The model (see Fig. 2) is composed of an input layer containing 7 neurons representing each of the 7 variables presented above. It turns out that when adding or removing both hemispheres of the brain left/right data as input, the results did not changed considerably. Therefore, the final model do not include those 4 attributes. The raw values were retained because the standardization and normalization processes of the data gave us poor results. After the input layer, there is the LSTM layer able to extract the sequential information useful for the prediction. The penultimate layer summarizes the information extracted in the lower layers in order to send it to the final layer that makes the final prediction. The activation function ReLU (Rectified Linear Unit) is used in all layers except for the last layer where the activation function is the Softmax. The loss is the binary cross entropy. The number of epoch was fixed to 50 and the batch size to 500. We used Adam as the optimizer. We implemented the models in python using Keras.

3.3 Results And Discussions : Can we predict Attention and Workload ?

The model that predict the attention gave 61% of accuracy and the model that predict the workload gave 76% of accuracy which is the average on 20 different training. Table 1 gives examples of predicted values vs real values of the two models. The second model performs well than the first model. These results show that it is possible to observe the variation of the attention and the workload of students during problem solving and predict how those mental states will vary in real time as students continue to solve problems. However, we noticed that the first model is slightly above the random model (better than chance) which means somehow that it is difficult to predict how the attention will vary across time compare to the workload. The model as designed is a general model in the sense that it does not take into account the specificities of each person.

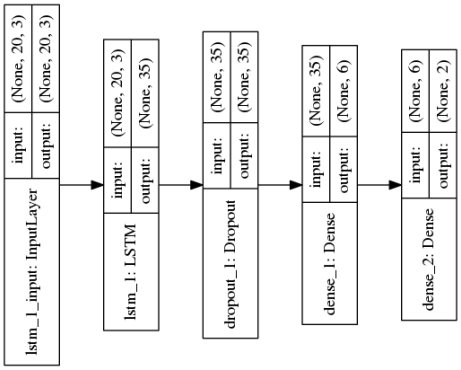

Fig. 2. Model to predict Attention and Workload in real time.

However, since people are different and may display different behaviours for the same task, it is important to incorporate a more contextual data to that model. One solution would be to add into these models a branch by using the attention mechanism [15] that would take as input specific attributes of the person whose attention or cognitive workload is to be predicted. The models need more contextual data to be able to figure out how the variables change across time. Nonetheless, our results seem promising, but are they meaningful ? EEG-based mental state detectors will presumably need to be much more accurate in the first place to help the design of accurate predictor and to help intelligent tutors in real time.

4 Conclusion

This preliminary study suggests that we can predict mental states variation across time. We found weak but above chance performance for predicting Attention. Even if the models built do not gave us good results, we have shown that it is possible to predict mental state across time based on past events. However the models need more contextual data to be able to predict more accurately the variations over time. As an example, we could include specific information on the student such as age, or the type of exercise they are currently solving. Our next step will be to first cluster students based on their behaviours and then to feed that new information to the models. The resulting solution will then be tested in a real-time setting.

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Table 1. Examples of prediction vs real data. [0, 1] for Attention means that the value to predict is above 0.6 and [1, 0] means that the value to predict is less than 0.6. [0, 1] for Workload means that the value to predict is above 0.02

Output	Real values	Predicted values
Attention	[0, 1]	[0.46971744, 0.5302825]
Attention	[1, 0]	[0.590336, 0.40963754]
Workload	[0, 1]	[0.495932, 0.504068]
Workload	[1, 0]	[0.9372175, 0.0627825]

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